Name: .... NEPTUN code: .... Major: ....

Probability Theory 1 exam, 6th January, 2026

Working hours: 100 min. A non-programmable calculator without internet connection can be used.
Maximal amount of points (with the Bonus): 110 points, but 100 points are already considered as 100%.

The. 1. (a) *(2+2 points)* Define the binomial distribution $\mathrm{BIN}(n,p)$ based on its descriptive meaning and derive from this the probabilities $\mathbb{P}(X=k)$ for the possible values of k if $X \sim \mathrm{BIN}(n,p)$.
(b) *(3+4 points)* Determine the expected value and variance of $\mathrm{BIN}(n,p)$.
(c) *(2+6 points)* State and prove the Poisson approximation theorem.

The. 2. (a) *(2 points)* Define the cumulative distribution function (c.d.f.) $F_X(x)$ of a random variable X.
(b) *(4+8 points)* State and prove the defining properties of the c.d.f.
(c) *(2 points)* Define the probability density function (p.d.f.) $f_X(x)$ of an absolutely continuous random variable X.
(d) *(3 points)* State the relation between the c.d.f. and the p.d.f. of an absolutely continuous random variable.

The. 3. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.
(a) *(2+2 points)* Define when we call two events $A, B \in \mathcal{F}$ independent. Define when we call a countable collection of events $A_1, A_2, \ldots$ independent.
(b) *(2 points)* Define when we call two random variables $X : \Omega \to \mathbb{R}$ and $Y : \Omega \to \mathbb{R}$ independent.
(c) *(3 points)* Give equivalent conditions for two random variables being independent (in the discrete case and the jointly absolutely continuous case)
(d) *(1+2 points)* Define the covariance $\mathrm{cov}(X,Y)$ of X and Y and calculate the covariance of a pair of independent, jointly absolutely continuous random variables X and Y.

Prac. 1. An archer is practicing shooting. The bull's eye (i.e. the center of his target) is the origin of the coordinate system, and the impact coordinates (X,Y) of his arrows are random variables such that $X \sim N(0, \frac{\sigma^2}{100})$, $Y \sim N(0, \frac{\sigma^2}{100})$, and X and Y are independent, where σ is the distance of the archer from the target in meters.
(a) *(4 points)* What is the probability that the archer hits the recrangle $[-2,2] \times [-1,1]$ if he stands exactly $\sigma = 10$ meters away from the target?
(b) *(7 points)* What is the probability that the archer hits the square $\{(x,y) : |x| + |y| \leq 2\}$ if he stands exactly $\sigma = 20$ meters away from the target?
(c) *(8 points)* If the target he aims is a disc of radius 1 meter (centered at the origin), what is the probability that the archer hits it as a function of σ?

Bonus: *(10 points)* In the previous exercise, derive the explicit formula for the expected Euclidean distance of the impact point of the arrow from the bull's eye as a function of σ.

Prac. 2. Let $X_1, X_2, \ldots$ and $Y_1, Y_2, \ldots$ be independent and identically distributed random variables with distribution $EXP(\lambda)$. Let $P_n = (\sum_{k=1}^n X_k, \sum_{k=1}^n Y_k)$ for $n = 1, 2, \ldots$ and let $\mathcal{S}$ be the following random set of random points on the plane: $\mathcal{S} = \{P_n : n = 1, 2, \ldots\}$. In other words: $\mathcal{S}$ consists of $(X_1, Y_1), (X_1 + X_2, Y_1 + Y_2), \ldots$
(a) *(4 points)* What is the probability that there are no points from $\mathcal{S}$ in the square $[0,t] \times [0,t]$?
(b) *(6 points)* What is the probab. that there are at least two points from $\mathcal{S}$ in the square $[0,t] \times [0,t]$?

Prac. 3. Santa's elves are in a hurry. There are 5 socks and 10 elves. Each elf throws one gift in a randomly chosen sock.
(a) *(6 points)* What is the expected value of the number of gifts in the first sock?
(b) *(7 points)* What is the expected value of the number of non-empty socks?
(c) *(8 points)* What is the variance of the number of non-empty socks?

$\Phi(z)$

z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000